Date:	
Program 1	Program to implement operations on Singly Linked List: (Insert, Delete, Traverse)

```
#include <malloc.h>
#include <stdio.h>
#include <stdlib.h>

typedef struct sllist {
int data;     struct
sllist *next;
} node;

node *head = NULL, *temp = NULL, *cur;

void create(); void
insert(); void
insert_beg(); void
insert_end(); void
insert_middle(); void
todelete();
void display();

int main() {     int choice = 0;     while (1) {
printf("\nSingly linked list operations");
printf("\n-----------------------------");
printf("\n1. Create");         printf("\n2.
Insertion");         printf("\n3. Traversal");
printf("\n4. Deletion");         printf("\n5.
Exit");         printf("\nEnter your choice: ");
       scanf("%d", &choice);


switch (choice) {
case 1:
create();
break;            case
2:                insert();
break;            case
3:
```

```
display();
break;          case
4:
todelete();
break;          case
5:
printf("\nEnd of the
list operations");
            exit(0);
default:
            printf("\nWrong choice---> Try again");
break;
        }
    }
}

void create() {
    int n, i;     node *cur;     printf("\nHow
many nodes to create: ");     scanf("%d",
&n);

    for (i = 1; i <= n; i++) {         temp =
(node*)malloc(sizeof(node));
printf("\nEnter data: ");         scanf("%d",
&temp->data);
        temp->next = NULL;

        if (head == NULL) {
head = temp;         } else
{
            cur = head;
            while (cur->next != NULL) {

cur = cur->next;
            }
            cur->next = temp;
        }
    }
}

void display() {     if (head
== NULL) {
printf("\nEmpty List");
    } else {
```

```
        cur = head;
printf("\nLinked list elements");
        printf("\n----------------------------");
while (cur != NULL) {
printf("\n%d", cur->data);
            cur = cur->next;
        }
        printf("\n---------------------");
    }
}

void todelete() {
int ele;
    node *prev = head;

    if (head == NULL) {         printf("\nEmpty list -->
Deletion is not possible");         return;
    }

    display();     printf("\nWhat
to delete? ");
    scanf("%d", &ele);

    cur = head;     while (cur->data != ele && cur-
>next != NULL) {         prev = cur;
        cur = cur->next;
    }

    if (cur->data == ele) {
if (cur == head) {             head
= cur->next;         } else {
prev->next = cur->next;
            free(cur);
        }
        if (cur->next == NULL) {             printf("\nElement
to be deleted is not in the list");
        }
    }
}

void insert() {     int option = 0;
temp = malloc(sizeof(node));
printf("\nEnter data: ");
scanf("%d", &temp->data);
```

```
   temp->next = NULL;

   printf("\nInsertion menu:");
printf("\n------------------");
printf("\n1. Beginning");
printf("\n2. Middle");     printf("\n3.
End");     printf("\nWhere to
insert??: ");
   scanf("%d", &option);

   switch(option) {
case 1:
insert_beg();
break;        case 2:
          if (head == NULL) {
printf("\nInsertion in middle is not possible");
return;            }            display();
insert_middle();            break;        case 3:
insert_end();            break;        default:
          printf("\nPlease enter the correct position");
   }
}

void insert_beg() {
temp->next = head;     head
= temp;
}

void insert_end() {     cur =
head;     while (cur->next !=
NULL) {
       cur = cur->next;
   }
   cur->next = temp;
}

void insert_middle() {
   int ele;     printf("\nAfter which
node??? ");
   scanf("%d", &ele);

   cur = head;     while (cur->data != ele && cur-
>next != NULL) {
       cur = cur->next;
```

```
    }

    if (cur->data == ele) {         temp-
>next = cur->next;
        cur->next = temp;
    }

    if (cur->next == NULL) {
printf("\n%d is not in the list", ele);
    }
}
```

```
"D:\40_Mohammed Anas DFS"

Singly Linked List Operations
------------------------------
1. Creation
2. Insertion
3. Display current List
4. Deletion
5. Search
6. Exit
Enter your choice: 1

How many nodes to create? 5
Enter data: 96
Enter data: 85
Enter data: 76
Enter data: 59
Enter data: 40

Singly Linked List Operations
------------------------------
1. Creation
2. Insertion
3. Display current List
4. Deletion
5. Search
6. Exit
Enter your choice: 3

Linked List Elements:
----------------------
96 -> 85 -> 76 -> 59 -> 40 -> NULL
```

```
Singly Linked List Operations
------------------------------
1. Creation
2. Insertion
3. Display current List
4. Deletion
5. Search
6. Exit
Enter your choice: 2

Enter data: 62

Insertion Menu
1. Beginning
2. Middle
3. End
Where to insert? 2

Linked List Elements:
----------------------
96 -> 85 -> 76 -> 59 -> 40 -> NULL

Insert after which element? 76

Inserted successfully

Singly Linked List Operations
------------------------------
1. Creation
2. Insertion
3. Display current List
4. Deletion
5. Search
6. Exit
Enter your choice: 3

Linked List Elements:
----------------------
96 -> 85 -> 76 -> 62 -> 59 -> 40 -> NULL
```

```
"D:\40_Mohammed Anas DFS"

Singly Linked List Operations
------------------------------
1. Creation
2. Insertion
3. Display current List
4. Deletion
5. Search
6. Exit
Enter your choice: 4

Linked List Elements:
----------------------
96 -> 85 -> 76 -> 62 -> 59 -> 40 -> NULL

Enter element to delete: 40

Element deleted successfully

Singly Linked List Operations
------------------------------
1. Creation
2. Insertion
3. Display current List
4. Deletion
5. Search
6. Exit
Enter your choice: 3

Linked List Elements:
----------------------
96 -> 85 -> 76 -> 62 -> 59 -> NULL
```

```
Singly Linked List Operations
------------------------------
1. Creation
2. Insertion
3. Display current List
4. Deletion
5. Search
6. Exit
Enter your choice: 5

Enter element to search: 76

Element 76 found at position 3

Singly Linked List Operations
------------------------------
1. Creation
2. Insertion
3. Display current List
4. Deletion
5. Search
6. Exit
Enter your choice: 6

End of operations
Press any key to continue . . .
```

Date:	
Program 2	Program to implement operations on Circular Linked List: insert, delete, traverse

```
#include <stdio.h>
#include <stdlib.h>

struct Node {     int
data;     struct Node
*next;
};

struct Node *head = NULL;

// -------- CREATE MULTIPLE NODES ---------- void
create() {
   int n, value;

   if (head != NULL) {
printf("List already created!\n");
      return;
   }

   printf("How many nodes do you want to enter? ");
scanf("%d", &n);

   for (int i = 1; i <= n; i++) {
printf("Enter data for node %d: ", i);
scanf("%d", &value);

      struct Node *newNode = malloc(sizeof(struct Node));         newNode-
>data = value;

      if (head == NULL) {
head = newNode;
         newNode->next = head;
      }
      else {
```

```
            struct Node *temp = head;
while (temp->next != head)
                temp = temp->next;

            temp->next = newNode;
            newNode->next = head;
        }
    }

    printf("List created with %d nodes.\n", n);
}

// -------- INSERT ---------- void insert()
{     if (head == NULL) {
printf("List empty. First create.\n");
return;
    }

    int choice, value, pos;     printf("\nInsert Menu:\n1. Beginning\n2. End\n3. Middle (after
position)\nEnter choice: ");     scanf("%d", &choice);

    printf("Enter data: ");
    scanf("%d", &value);

    struct Node *newNode = malloc(sizeof(struct Node));     newNode-
>data = value;

    // Insert at beginning     if
(choice == 1) {         struct Node
*temp = head;         while
(temp->next != head)
            temp = temp->next;

        temp->next = newNode;         newNode-
>next = head;
        head = newNode;

        printf("Inserted at beginning.\n");
    }

    // Insert at end     else if
(choice == 2) {         struct Node
*temp = head;         while
(temp->next != head)
```

```
            temp = temp->next;

        temp->next = newNode;
        newNode->next = head;

        printf("Inserted at end.\n");
    }

    // Insert middle     else if (choice == 3) {
printf("Insert after which position? ");
scanf("%d", &pos);

        struct Node *temp = head;         for
(int i = 1; i < pos; i++) {             temp =
temp->next;             if (temp == head) {
printf("Position out of range.\n");
                return;
            }
        }

        newNode->next = temp->next;
        temp->next = newNode;

        printf("Inserted after position %d.\n", pos);
    }

    else {
        printf("Invalid choice.\n");
    }
}

// -------- TRAVERSAL ----------
void traversal() {     if (head ==
NULL) {         printf("List
empty.\n");
        return;
    }

    struct Node *temp = head;
    printf("Circular Linked List: ");

    do {         printf("%d ",
temp->data);         temp =
temp->next;
```

```
    } while (temp != head);

    printf("\n");
}
```

```
// -------- DELETE BY VALUE ONLY ----------
void deletion() {     if (head == NULL) {
printf("List empty.\n");
        return;
    }

    int value;     printf("Enter value
to delete: ");
    scanf("%d", &value);

    struct Node *temp = head, *prev = NULL;

    // If deleting head node
if (head->data == value) {
        struct Node *last = head;

        while (last->next != head)
            last = last->next;

        // only 1 node         if
(head->next == head) {
free(head);             head =
NULL;         } else {
last->next = head->next;
temp = head;             head =
head->next;             free(temp);
        }

        printf("Value %d deleted.\n", value);
        return;
    }

    // Search in remaining nodes
    do {         prev
= temp;
        temp = temp->next;

        if (temp->data == value) {
prev->next = temp->next;
free(temp);             printf("Value %d
deleted.\n", value);             return;
        }

    } while (temp != head);
```

```
   printf("Value not found.\n");
}

// -------- MAIN ---------- int
main() {
   int choice;

   while (1) {
      printf("\n--- Circular Linked List Menu ---\n");        printf("1.
Create\n2. Insert\n3. Traversal\n4. Delete \n5. Exit\n");        printf("--
--------------------------------\nEnter your choice: ");
      scanf("%d", &choice);

      switch (choice) {            case 1:
create(); break;            case 2: insert();
break;            case 3: traversal(); break;
case 4: deletion(); break;            case 5:
exit(0);            default: printf("Invalid
choice!\n");
      }
   }

   return 0;
}
```

```
 Circular linked list operations
-----------------------
 1. Create
 2. Insertion
 3. Traversal
 4. Deletion
 5. Exit
 Enter your choice: 1

 how many nodes to create: 3
Enter data: 15
Enter data: 25
Enter data: 35

 Circular linked list operations
-----------------------
 1. Create
 2. Insertion
 3. Traversal
 4. Deletion
 5. Exit
 Enter your choice: 3

 List: 15 25 35
```

```
 Circular linked list operations
-----------------------
 1. Create
 2. Insertion
 3. Traversal
 4. Deletion
 5. Exit
 Enter your choice: 2

 Insertion menu
 1. Beginning
 2. Middle
 3. End
 where to insert: 2
Enter data: 45

 List: 15 25 35
 After which node? 25

 Circular linked list operations
-----------------------
 1. Create
 2. Insertion
 3. Traversal
 4. Deletion
 5. Exit
 Enter your choice: 3

 List: 15 25 45 35
```

```
 Circular linked list operations
-----------------------
 1. Create
 2. Insertion
 3. Traversal
 4. Deletion
 5. Exit
 Enter your choice: 4

 Enter element to delete: 45

 Circular linked list operations
-----------------------
 1. Create
 2. Insertion
 3. Traversal
 4. Deletion
 5. Exit
 Enter your choice: 3

 List: 15 25 35
 Circular linked list operations
-----------------------
 1. Create
 2. Insertion
 3. Traversal
 4. Deletion
 5. Exit
 Enter your choice: 5

 End of operationPress any key to continue . . .
```

Date:	
Program 3	Program to implement stack operations using pointers (push, pop, process)

```
#include <stdio.h>

#include <conio.h> #include

<malloc.h> typedef struct

stack

{

        int data;

struct stack *next; }node;

node *top=NULL, *temp;
```

```
void push(int); int pop(); void display();
void main() {  int choice=0, item, i, n,
ele[100];
        while (1)
        {
                printf("\n Stack Operations");
        printf("\n -----------------");
printf("\n 1.push");
                printf("\n 2.pop");
printf("\n 3.Display");
printf("\n 4.Exit");             printf("\n Enter
your choice: ");                scanf("%d",
&choice);               switch(choice)
                {
                        case 1:printf("\nEnter the number of items to push: ");
                        scanf("%d", &n);
                        printf("\nEnter the elements: ");
                        for (i=0;i<n;i++)
                        {
                        scanf("%d", &ele[i]);
        push(ele[i]);
                        }
                        break;
```

```
                    case 2:if (top==NULL)
            printf("\n Empty Stack..!");
                    else
                            printf("\nItem popped: %d", pop());
                    break;


                    case 3:display();
                    break;

                    case 4:printf("\nEnding Stack Operations...");
                    getch();
        exit(10);
                  default: printf("\n Invalid choice..!");
            }
        }
} void push(int
item)
{

temp=malloc(sizeof(node));
temp->data=item;      temp-
>next=NULL;           temp-
>next=top;      top=temp;
```

```
} int
pop()
{
        int data;
temp=top;
top=temp->next;
data=temp->data;
free(temp);     return
data;
}
void display()
{
        if (top==NULL)
        printf("\nEmpty stack..!");
        else
        {
                printf("\nStack members");                printf("\n------
---------");
                temp=top;
while(temp!=NULL)
                {
                        printf("\n%d", temp->data);
                        temp=temp->next;
                }
```

```
                printf("\n-----------------------");

        }

}
```

```
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 Stack Operation
 ---------------
 1. Push
 2. Pop
 3. Display
 4. Exit
 Enter your Choice:1

 Enter the number of items to push:5

 Enter the Elements:16
34
68
94
71

 Stack Operation
 ---------------
 1. Push
 2. Pop
 3. Display
 4. Exit
 Enter your Choice:3

 Stack Members
--------------
 71
 94
 68
 34
 16
 ----------
```

```
 Stack Operation
 ---------------
 1. Push
 2. Pop
 3. Display
 4. Exit
 Enter your Choice:2

 Popped item =71
 Stack Operation
 ---------------
 1. Push
 2. Pop
 3. Display
 4. Exit
 Enter your Choice:2

 Popped item =94
 Stack Operation
 ---------------
 1. Push
 2. Pop
 3. Display
 4. Exit
 Enter your Choice:3

 Stack Members
--------------
 68
 34
 16
 ----------
```

```
Stack Operation
---------------
1. Push
2. Pop
3. Display
4. Exit
Enter your Choice:5

invalid choice
Stack Operation
---------------
1. Push
2. Pop
3. Display
4. Exit
Enter your Choice:4

Ending Stack Operations
```

Date:	
Program 4	Program to implement conversion of infix expression to its postfix form

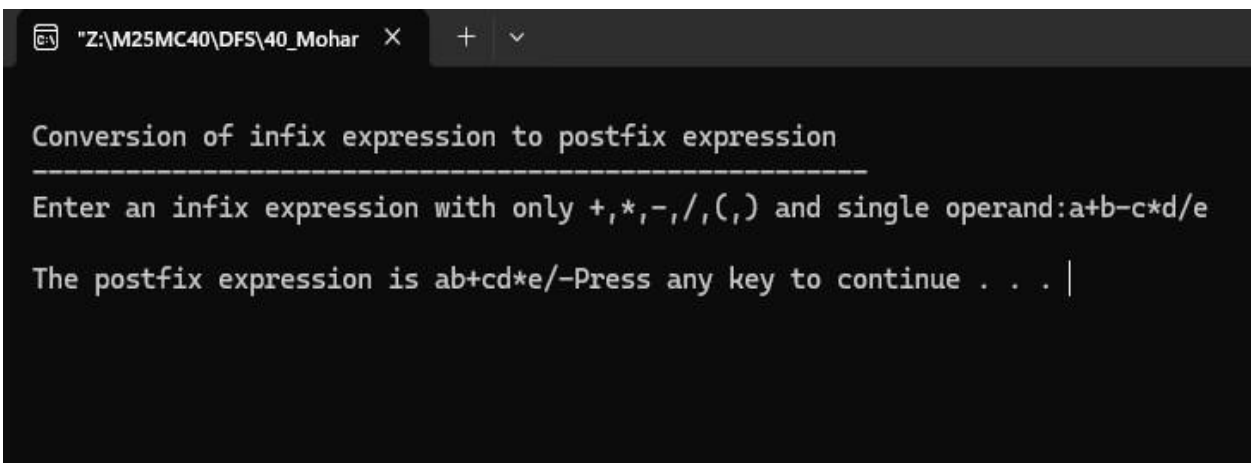
```
#include<stdio.h>

#include<string.h>

#include<malloc.h>


typedef struct opstack

{     char data;     struct

opstack *next;

} node;


node *TOP=NULL,*temp=NULL;
```

```
void push(char); char
pop(); int
stackpriority(char); int
infixpriority(char);

int main() {     char
infix[50],postfix[50],ch,ch1;     int
i=0,j=0;

   printf("\n Conversion of infix expression to postfix expression");     printf("\n --
-----------------------------------------------------");     printf("\n Enter an infix
expression with only +,*,-,/,(,) and single operand: \n");     gets(infix);

   strcpy(postfix," ");

   while(infix[i]!='\0')
   {        ch=infix[i];        if(('A'<=ch &&
ch<='Z')||('a'<=ch && ch<='z'))
postfix[j++]=ch;

       else if(ch=='(')
       {
push(ch);
       }        else
if(ch==')')
```

```
        {           ch1=pop();
while(ch1!='(' && ch1!='\0')
            {
postfix[j++]=ch1;
ch1=pop();
            }
        }
else
        {
            while(TOP!=NULL && stackpriority(TOP->data) >= infixpriority(ch))
            {
postfix[j++] = pop();
            }
push(ch);
        }
i++;
    }

    while(TOP!=NULL)
    {       ch1=pop();
postfix[j++]=ch1;
    }

    postfix[j]='\0';     printf("\n The postfix expression
is %s \n",postfix);
```

```
}


void push(char ch)
{     node *temp;

temp=(node*)malloc(sizeof(node));

temp->data=ch;     temp->next=TOP;

   TOP=temp;

}


char pop() {

node *temp;

char x;


   if(TOP==NULL)

       return '\0';


   temp=TOP;     x=temp-

>data;     TOP=TOP->next;

   free(temp);

return x; }


int infixpriority(char ch)

{
   if(ch=='+' || ch== '-')

return 1;     else if(ch=='*'

|| ch=='/')        return 2;
```

```
else if(ch=='(')        return
3;     else        return 0; }


int stackpriority(char ch) {
if(ch=='+' || ch=='-')
return 1;     else if(ch=='*'
|| ch=='/')        return 2;
else if(ch=='(')        return
0;          else        return
100;     }
```



```
Conversion of infix expression to postfix expression
-----------------------------------------------------
Enter an infix expression with only +,*,-,/,(,) and single operand:
(a*b)*(f+d)/c-f

The postfix expression is ab*fd+*c/f-
```

```
Conversion of infix expression to postfix expression
-----------------------------------------------------
Enter an infix expression with only +,*,-,/,(,) and single operand:
a/b-(c*d)+e/f+g*h

The postfix expression is ab/cd*-ef/+gh*+
```

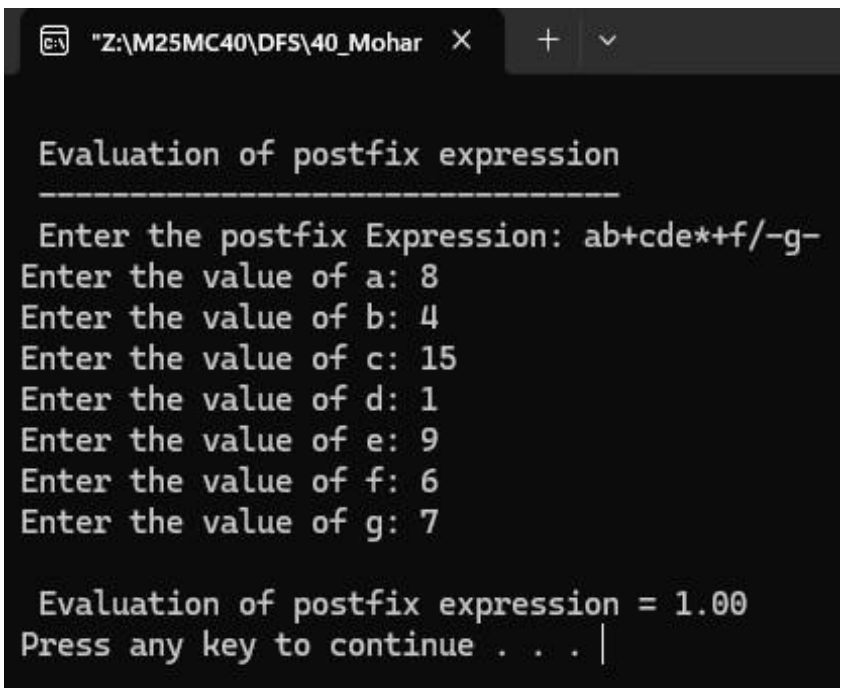
Date:	
Program 5	Program to implement evaluation of postfix expression

```
#include <stdio.h>

#include<conio.h>

#include<string.h>

#include<malloc.h>


typedef struct opstack

{

        char data;      struct

opstack *next;
```

```
}node; node *top=NULL,
*temp=NULL;

void main()
{
        char postfix[50],
ch;     void push(int);
int pop();
        int chval, res, v1, v2, i=0, item;
printf("\n evaluation of postfix expression");
printf("\n-------------------------------------");
printf("\nEnter the postfix expression: ");
gets(postfix); while(postfix[i]!='\0')
        {
                ch=postfix[i];
                if(('A'<=ch&&ch<='Z')||('a'<=ch&&ch<='z'))
                {
                        printf("\n Enter a value of %c: ",ch);
                        scanf("%d",
&chval);                        fflush(stdin);
                push(chval);            }else
                {
```

```
                        v2=pop();
        v1=pop();
switch(ch)
                        {
                                case '+': res=v1+v2;
                                break;
                                case '-': res=v1-v2;
                                break;
                                case '*': res=v1*v2;
                                break;
                                case '/': res=v1/v2;
                                break;
                        }
                        push(res);
                }
                i++;
        }
        printf("\n Evaluated value of postfix expression %d", pop());
printf("\n");
} void push(int
item)
{
```

```
        temp=malloc(sizeof
(node));        temp->data=item;
temp->next=top;        top=temp;
} int
pop()
{
        temp=top;
top=temp->next;        return
(temp->data);
}
```



```
 evaluation of postfix expression
-----------------------------------
Enter the postfix expression: ab/cd+f*eg*

 Enter a value of a: 20

 Enter a value of b: 2

 Enter a value of c: 6

 Enter a value of d: 14

 Enter a value of f: 2

 Enter a value of e: 5

 Enter a value of g: 10

 Evaluated value of postfix expression 50
```

Date:	
Program 6	Program to implement various operations on queue using pointers (Insertion, Deletion, Process the Queue)

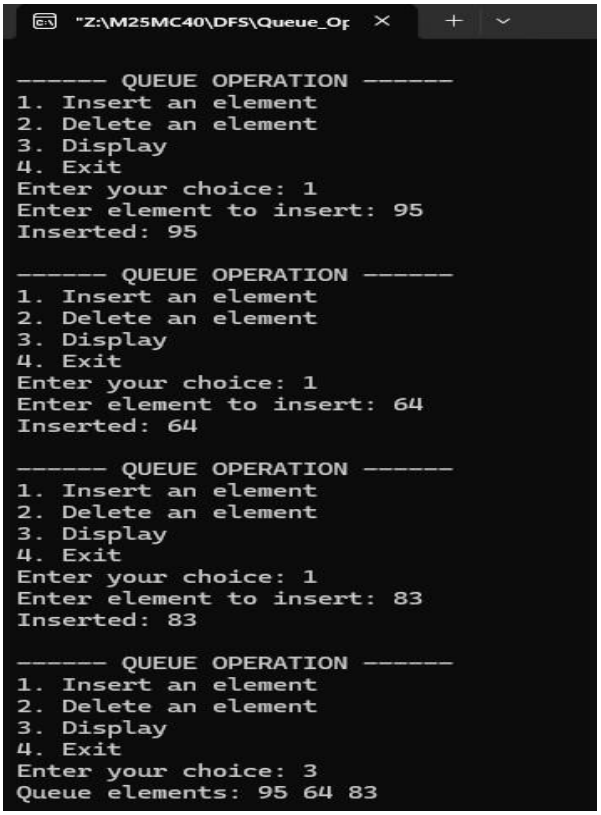
```
#include<stdio.h>
#include<conio.h>
#include<malloc.h>
void qinsert(); void
qdelete(); void
qdisplay(); typedef
struct queue
{
        int data;
struct queue *next;
```

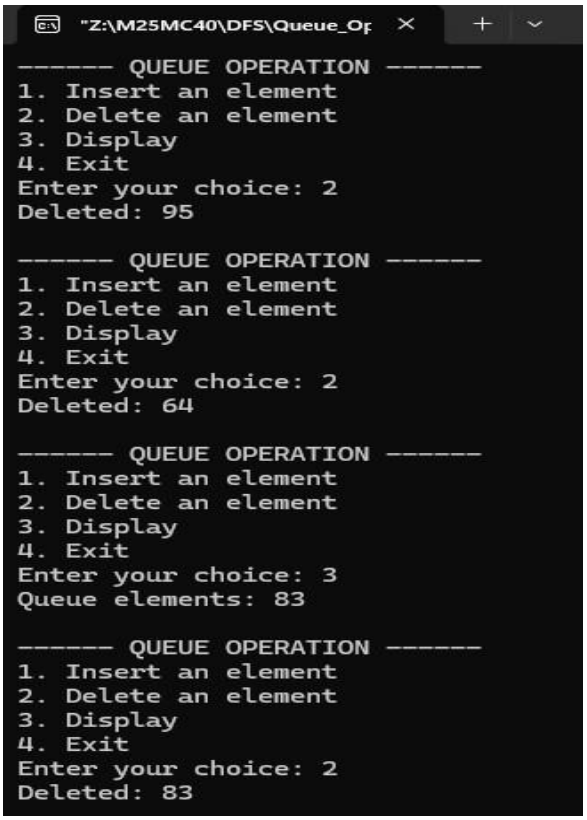
```
}node; node *front=NULL, *rear=NULL,
*temp; void main()
{
        int choice=0;
while(1)
        {
                printf("\nQueue Operations")
                printf("\n----------------");
                printf("\n1.Insertion");
                printf("\n2.Deletion"); printf("\n3.Display");
                printf("\n4.Exit");
                printf("Enter your choice:");
                scanf("%d",&choice);
switch(choice)
        {
                case 1:qinsert();
                break;
                case 2:qdelete();
                break;
                case 3:qdisplay();
                break;
                case 4:printf("\nEnd of the operation");
                exit(0);
                default: printf("\nWrongf choice");
```

```
        }
} } void
qinsert()
{
        int n,i;
        printf("Enter the number of nodes to insert: ");
        scanf("%d",&n);
for (i=0;i<n;i++)
        {

temp=malloc(sizeof(node));
printf("Enter data: ");
scanf("%d",&temp->data);
temp->next=NULL;   if
(rear==NULL)
        {
                rear=temp;
front=temp;
        }
        else
                {
                        rear->next=temp;
        rear=temp;
                }
```

```
        }
} void
qdelete()
{
        if (front==NULL)
        {
                printf("\nEmpty queue->deletion not possible");
        }
        else{
temp=front;
                printf("\nDeleted item is %d",front->data);              front=front-
>next;                  free(temp);


                if (front==NULL)
                {
                        rear=NULL;
                }
        }
} void
qdisplay()
{if (front==NULL)
        {
                printf("\nEmpty queue");
        }else
```

```
  {
            temp=front;
      printf("\nQueue elements");
                  printf("\n------------------\n");
                  while(temp!=NULL)      {
         printf("%d\n",temp->data);
temp=temp->next;
                  }
         }
}
```





```
------ QUEUE OPERATION ------
1. Insert an element
2. Delete an element
3. Display
4. Exit
Enter your choice: 2
Deleted: 83

------ QUEUE OPERATION ------
1. Insert an element
2. Delete an element
3. Display
4. Exit
Enter your choice: 3
Queue is empty!

------ QUEUE OPERATION ------
1. Insert an element
2. Delete an element
3. Display
4. Exit
Enter your choice: 5
Invalid choice! Try again.

------ QUEUE OPERATION ------
1. Insert an element
2. Delete an element
3. Display
4. Exit
Enter your choice: 4
Program exiting...
Press any key to continue . . .
```

Date:	
Program 7	Create a double-ended queue (Dequeue) with the following operations: (a) insert in the Beginning (b) insert in the end (c) delete from beginning (d) delete from end

```
#include <stdio.h>

#define MAX 5


int deque[MAX]; int

front=-1, rear=-1;


/*FUNCTION PROTOTYPES*/

void insertfront(int);

void insertrear(int);

void deletefront();

void deleterear(); void

display();


void main(){    int

choice, item;

        do

        {

                printf("\n DOUBLE ENDED QUEUE");

                printf("\n ----------------");

                printf("\n 1.Insert in the front.");
```

```
                printf("\n 2.Insert in the rear.");
                printf("\n 3.Delete from front.");
                printf("\n 4.Delete from rear.");
                printf("\n 5.Display.");
                printf("\n 6.Exit.\n");

                printf("Enter your choice: ");
scanf("%d", &choice);          switch(choice)
        {
                case 1:
                printf("Enter the element to insert in the front:");
                scanf("%d", &item);
insertfront(item);
                break;
case 2:
                printf("Enter the element to insert in the rear:");
                scanf("%d", &item);
insertrear(item);
                break;
case 3:
                deletefront();
                break;
                case 4:
```

```
                deleterear();

                break;
            case 5:

                display();

break;          case 6:

                printf("Exiting the program!\n");

                break;

default:

                printf("Invalid choice..");

        }


        }while (choice!=6);

}


/*INSERT IN FRONT*/

void insertfront(int item)

{

        if((front==0&&rear==MAX-1)||(front==rear+1))

        {

                printf("Deque Overflow!");

                return;

        }

        if (front==-1)

        {
```

```
            front=rear=0;
        }
        else if(front==0)
        {
            front=MAX-1;
        }
        else
        {
            front--;
        }
        deque[front]=item;
}

/*INSERT IN REAR*/
void insertrear(int item)
{
        if ((front==0&&rear==MAX-1)||(front==rear+1))
        {
            printf("Deque Overfow\n");
            return;
        }
        if (rear==-1)
        {
            front=rear=0;
```

```
        } else if(rear==MAX-
        1)
        {
                rear=0;
        }
        else
        {
                rear++;
        }
        deque[rear]=item;
} void
deletefront()
{
        if (front==-1)
        {
                printf("Deque Overflow\n");
                return;
        }
 printf("Deleted element: %d\n", deque[front]);  if
(front==rear)
        {
                front=rear=-1;
        }
        else if(front==MAX-1)
        {
```

```
                front=0;
        }
        else
        {

                front++;
        } } void
deleterear()
{
        if (rear==-1)
        {
                printf("Deque Underflow\n");
                return;
        }
 printf("Deleted element :%d\n",deque[rear]);  if
(front==rear)
        {
                front=rear=-1;
        }
        else if(rear==0)
        {
                rear=MAX-1;
        }
        else{
rear--;
```

```
        }
}
void display()
{
        int i;    if
(front==-1)
        {
                printf("Deque is empty\n");
                return;
        }
        printf("Deque elements: \t");
        i=front;
while(1)
        {
                printf("\n%d\t",deque[i]);
        if (i==rear)
                break;
i=(i+1)%MAX;
        }
}
```

```
"Z:\M25MC40\DFS\dq.exe"

----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 1
Enter element to insert at front: 15


----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 1
Enter element to insert at front: 25
```

```
"Z:\M25MC40\DFS\dq.exe"

----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 2
Enter element to insert at rear: 55


----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 2
Enter element to insert at rear: 45


----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 5
Deque elements: 25      15      55      45
```

```
----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 3
Deleted element: 25


----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 4
Deleted element: 45
```

```
----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 5
Deque elements: 15      55

----- DOUBLE ENDED QUEUE (DEQUE) -----
1. Insert at Front
2. Insert at Rear
3. Delete from Front
4. Delete from Rear
5. Display
6. Exit
Enter your choice: 6
Exiting program.
Press any key to continue . . .
```

Date:	
Program 8	Implement sort algorithms (Quick sort, Insertion sort)

```
#include <stdio.h>
#include <stdlib.h>

void insertionSort(int arr[], int n) {
    int i, key, j;
    for (i = 1; i < n; i++) {
        key = arr[i];
        j = i - 1;
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j = j - 1;
        }
        arr[j + 1] = key;
    }
}
void swap(int* a, int* b) {
    int t = *a;
    *a = *b;
    *b = t;
}
int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = (low - 1);
    int j;
    for (j = low; j <= high - 1; j++) {
```

```
        if (arr[j] < pivot) {
            i++;
            swap(&arr[i], &arr[j]);
  }
    }
    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}
void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}
void printArray(int arr[], int n) {
    int i;
    for (i = 0; i < n; i++)
        printf("%d ", arr[i]);
    printf("\n");
}
int main() {
    int n, i, choice;
    int arr[100];
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    printf("\n--- Sorting Menu ---\n");
```

```
    printf("1. Insertion Sort\n");
    printf("2. Quick Sort\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            insertionSort(arr, n);
            printf("Sorted array using Insertion Sort: \n");
            break;
        case 2:
            quickSort(arr, 0, n - 1);
            printf("Sorted array using Quick Sort: \n");
            break;




        default:
            printf("Invalid choice!\n");
            return 0;
    }
    printArray(arr, n);
    return 0;
}
```

```
"D:\DFS Lab Programs\Inserti

Enter number of elements: 5
Enter 5 elements:
85 6 36 79 42

--- Sorting Menu ---
1. Insertion Sort
2. Quick Sort
Enter choice: 1
Sorted array using Insertion Sort:
6 36 42 79 85
Press any key to continue . . .
```

```
"D:\DFS Lab Programs\Inserti

Enter number of elements: 5
Enter 5 elements:
100 41 28 6 81

--- Sorting Menu ---
1. Insertion Sort
2. Quick Sort
Enter choice: 2
Sorted array using Quick Sort:
6 28 41 81 100
Press any key to continue . . .
```

Date:	
Program 9	Program to Implement Linear Search & Binary Search

```
#include <stdio.h>

#include <stdlib.h>


int linearSearch(int arr[], int n, int key) {

    int i;

    for (i = 0; i < n; i++) {

        if (arr[i] == key) {

            return i;

        }

    }

    return -1;

}


int binarySearch(int arr[], int n, int key) {

    int low = 0, high = n - 1;

    while (low <= high) {

        int mid = low + (high - low) / 2;


        if (arr[mid] == key)
```

```
            return mid;   // ? fixed here


        if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}


int main() {
    int n, i, key, choice, result;
    int arr[100];


    printf("Enter number of elements: ");
    scanf("%d", &n);


    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }


    printf("\n--- Search Menu ---\n");
```

```
printf("1. Linear Search\n");
printf("2. Binary Search (Ensure array is sorted!)\n");
printf("Enter choice: ");
scanf("%d", &choice);

printf("Enter element to search: ");
scanf("%d", &key);

switch (choice) {
    case 1:
        result = linearSearch(arr, n, key);
        break;
    case 2:
        result = binarySearch(arr, n, key);
        break;
    default:
        printf("Invalid choice!\n");
        return 0;
}
```

```
    if (result != -1)
        printf("Element found at index: %d (Position: %d)\n", result, result + 1);
    else
        printf("Element not found in the array.\n");


    return 0;
}
```

```
"D:\DFS Lab Programs\Linear_

Enter number of elements: 5
Enter 5 elements:
65 46 2 97 350

--- Search Menu ---
1. Linear Search
2. Binary Search (Ensure array is sorted!)
Enter choice: 1
Enter element to search: 2
Element found at index: 2 (Position: 3)
Press any key to continue . . .
```

```
"D:\DFS Lab Programs\Linear_

Enter number of elements: 6
Enter 6 elements:
25 6 78 650 35 10

--- Search Menu ---
1. Linear Search
2. Binary Search (Ensure array is sorted!)
Enter choice: 1
Enter element to search: 650
Element found at index: 3 (Position: 4)
Press any key to continue . . .
```

```
"D:\DFS Lab Programs\Linear_

Enter number of elements: 5
Enter 5 elements:
1 46 58 77 100

--- Search Menu ---
1. Linear Search
2. Binary Search (Ensure array is sorted!)
Enter choice: 2
Enter element to search: 77
Element found at index: 3 (Position: 4)
Press any key to continue . . .
```

```
"D:\DFS Lab Programs\Linear_

Enter number of elements: 4
Enter 4 elements:
108 119 165 250

--- Search Menu ---
1. Linear Search
2. Binary Search (Ensure array is sorted!)
Enter choice: 2
Enter element to search: 108
Element found at index: 0 (Position: 1)
Press any key to continue . . .
```

Date:	
Program 10	Program to demonstrate operations on Binary Trees: (Create, Insert, Delete, Process)

```
#include <stdio.h>

#include <stdlib.h>

struct Node {

    int data;

    struct Node *left, *right;

};

struct Node* createNode(int data) {

    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

    newNode->data = data;

    newNode->left = newNode->right = NULL;

    return newNode;

}

void inorder(struct Node* temp) {

    if (temp == NULL) return;

    inorder(temp->left);

    printf("%d ", temp->data);

    inorder(temp->right);

}

struct Node* insert(struct Node* root, int data) {
```

```
    if (root == NULL) {
        return createNode(data);
    }
    struct Node* queue[100];
    int front = 0, rear = 0;
    queue[rear++] = root;
    while (front < rear) {
        struct Node* temp = queue[front++];
 if (temp->left == NULL) {
            temp->left = createNode(data);
            break;
        } else {
            queue[rear++] = temp->left;
        }
        if (temp->right == NULL) {
            temp->right = createNode(data);
            break;
        } else {
            queue[rear++] = temp->right;
        }
    }
    return root;
}
```

```
void deleteDeepest(struct Node* root, struct Node* d_node) {
    struct Node* queue[100];
    int front = 0, rear = 0;
    queue[rear++] = root;
    while (front < rear) {
        struct Node* temp = queue[front++];
        if (temp == d_node) {
            temp = NULL;
            free(d_node);
            return;
        }
        if (temp->right) {
            if (temp->right == d_node) {
                temp->right = NULL;
                free(d_node);
                return;
            } else queue[rear++] = temp->right;
      }
        if (temp->left) {
            if (temp->left == d_node) {
                temp->left = NULL;
                free(d_node);
                return;
```

```
            } else queue[rear++] = temp->left;
        }
    }
}
struct Node* deletion(struct Node* root, int key) {
    if (root == NULL) return NULL;
    if (root->left == NULL && root->right == NULL) {
        if (root->data == key) return NULL;
        else return root;
    }
    struct Node* queue[100];
    int front = 0, rear = 0;
    queue[rear++] = root;
    struct Node* temp;
    struct Node* key_node = NULL;
    while (front < rear) {
        temp = queue[front++];
        if (temp->data == key) key_node = temp;
        if (temp->left) queue[rear++] = temp->left;
        if (temp->right) queue[rear++] = temp->right;
    } if (key_node != NULL) {
        int x = temp->data;
        deleteDeepest(root, temp);
```

```
        key_node->data = x;
    } return root;
}
int main() {
    struct Node* root = NULL;
    int choice, val,n,i;
    while (1) {
        printf("\n1. Insert\n2. Delete\n3. Display (Inorder)\n4. Exit\n");
        printf("Choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
             printf("enter no of elements to insert:\t");
             scanf("%d",&n);
             for(i=0;i<n;i++){
              printf("Value to insert:");
                 scanf("%d", &val);
                 root = insert(root, val);
             }
               break;
            case 2:
               printf("Value to delete: ");
               scanf("%d", &val);
```

```
            root = deletion(root, val);
            break;
        case 3:
            printf("Tree: ");
            inorder(root);
            printf("\n");
            break;
        case 4:
            exit(0);
    }
}
  return 0;
}
```

```
1. Insert
2. Delete
3. Display (Inorder)
4. Exit
Choice: 3
Tree: 10 98 64 51

1. Insert
2. Delete
3. Display (Inorder)
4. Exit
Choice: 4
Press any key to continue . . .
```

```
1. Insert
2. Delete
3. Display (Inorder)
4. Exit
Choice: 3
Tree: 10 98 33 64 51

1. Insert
2. Delete
3. Display (Inorder)
4. Exit
Choice: 2
Value to delete: 33
```

```
"D:\DFS Lab Programs\Binary

1. Insert
2. Delete
3. Display (Inorder)
4. Exit
Choice: 1
enter no of elements to insert: 5
Value to insert:64
Value to insert:98
Value to insert:51
Value to insert:10
Value to insert:33
```

Date:	
Program 11	Program to implement Binary Search Tree operations: Insert, Traversals (Inorder, Preorder, Postorder)

```
#include <stdio.h>

#include <stdlib.h>

struct Node {

    int data;

    struct Node *left, *right;

};

struct Node* createNode(int value) {

    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

    newNode->data = value;

    newNode->left = newNode->right = NULL;

    return newNode;

}

struct Node* insert(struct Node* root, int value) {

    if (root == NULL) {

        return createNode(value);

    }

    if (value < root->data) {

        root->left = insert(root->left, value);

    } else if (value > root->data) {
```

```
        root->right = insert(root->right, value);
    }
    return root;
}
void inorder(struct Node* root) {
    if (root != NULL) {
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
            }
}
void preorder(struct Node* root) {
    if (root != NULL) {
        printf("%d ", root->data);
        preorder(root->left);
        preorder(root->right);
    }
}
void postorder(struct Node* root) {
    if (root != NULL) {
        postorder(root->left);
        postorder(root->right);
        printf("%d ", root->data);
```

```
    }
}
int main() {
    struct Node* root = NULL;
    int choice, value;


    printf("--- Binary Search Tree Operations ---\n");
    while(1) {
        printf("\n1. Insert\n2. Inorder\n3. Preorder\n4. Postorder\n5. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);


        switch(choice) {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &value);
                root = insert(root, value);
                break;
                  case 2:
                printf("Inorder Traversal: ");
                inorder(root);
                printf("\n");
                break;
```

```
        case 3:
            printf("Preorder Traversal: ");
            preorder(root);
            printf("\n");
            break;
        case 4:
            printf("Postorder Traversal: ");
            postorder(root);
            printf("\n");
            break;
        case 5:
            exit(0);
        default:
            printf("Invalid choice!\n");
    }
  }
  return 0;
}
```

```
"D:\DFS Lab Programs\BST_O

--- Binary Search Tree Operations

1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 1
Enter value to insert: 3

1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 1
Enter value to insert: 12

1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 1
Enter value to insert: 46
```

```
1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 1
Enter value to insert: 67

1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 1
Enter value to insert: 90
```

```
1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 2
Inorder Traversal: 3 12 46 67 90

1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 3
Preorder Traversal: 3 12 46 67 90

1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 4
Postorder Traversal: 90 67 46 12 3

1. Insert
2. Inorder
3. Preorder
4. Postorder
5. Exit
Enter choice: 5
Press any key to continue . . .
```

Date:	
Program 12	Program to display the Adjacency Matrix of a Graph

```
#include <stdio.h>

#define MAX 100


int main()

{

    int adj[MAX][MAX];

    int vertices, edges;

    int src, dest,i,j;

    printf("Enter the number of vertices (max 100): ");

    scanf("%d", &vertices);

    printf("Enter the number of edges: ");

    scanf("%d", &edges);

    for (i = 0; i < vertices; i++) {

        for (j = 0; j < vertices; j++) {

            adj[i][j] = 0;

        }

    }

    printf("\nEnter edges (source destination):\n");
```

```
    for (i = 0; i < edges; i++) {
        printf("Edge %d: ", i + 1);
        scanf("%d %d", &src, &dest);
        adj[src][dest] = 1;
    }
    printf("Adjacency Matrix:\n");
    printf("    ");
    for (i = 0; i < vertices; i++) {
      printf(" v%d ", i);
    }
    printf("\n");
    for (i = 0; i < vertices; i++)
    {
        printf("v%d  ", i);
        for (j = 0; j < vertices; j++)
       {
            printf(" %d  ", adj[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

```
"D:\DFS Lab Programs\G_Adja

Enter the number of vertices (max 100): 6
Enter the number of edges: 7

Enter edges (source destination):
Edge 1: 2
3
Edge 2: 4
2
Edge 3: 5
6
Edge 4: 1
3
Edge 5: 6
4
Edge 6: 7
8
Edge 7: 7
5
```

```
Adjacency Matrix:
     v0  v1  v2  v3  v4  v5
v0   0   0   0   0   0   0
v1   0   0   0   1   0   0
v2   0   0   0   1   0   0
v3   0   0   0   0   0   0
v4   0   0   1   0   0   0
v5   0   0   0   0   0   0
Press any key to continue . . .
```

Date:	
Program 13 (a)	Program to perform Depth First Traversal (DFS) on a Graph

```
#include <stdio.h>

#define MAX 100


int adj[MAX][MAX];

int visited[MAX];

int vertices;

void DFS(int vertex) {

 int i;

    printf("%d ", vertex);

    visited[vertex] = 1;

    for (i = 0; i < vertices; i++) {

        if (adj[vertex][i] == 1 && !visited[i]) {

            DFS(i);

   } }}

void displayMatrix() {

 int i,j;

    printf("Adjacency Matrix:\n");

    printf("     ");
```

```
    for (i = 0; i < vertices; i++) {
        printf(" v%d ", i);
    }


    printf("\n");
    for (i = 0; i < vertices; i++) {
        printf("v%d   ", i);
        for (j = 0; j < vertices; j++) {
            printf(" %d  ", adj[i][j]);
        }
          printf("\n");
    }
}
int main() {
    int edges, src, dest, start,i,j;
    printf("Enter the number of vertices (max 100): ");
    scanf("%d", &vertices);
    printf("Enter the number of edges: ");
    scanf("%d", &edges);
    for (i = 0; i < vertices; i++) {
        for (j = 0; j < vertices; j++) {
            adj[i][j] = 0;
        }
```

```
        visited[i] = 0;
    }
    printf("\nEnter edges (source destination):\n");
    for (i = 0; i < edges; i++) {
        printf("Edge %d: ", i + 1);
        scanf("%d %d", &src, &dest);
        adj[src][dest] = 1;
    }
    displayMatrix();
    printf("\nEnter starting vertex for DFS: ");
    scanf("%d", &start);
    for (i = 0; i < vertices; i++) {
        visited[i] = 0;
    }
    printf("\nDFS Traversal starting from vertex %d: ", start);
    DFS(start);
    printf("\n");
    return 0;
}
```

```
"D:\DFS Lab Programs\Depth_

Enter the number of vertices (max 100): 7
Enter the number of edges: 8

Enter edges (source destination):
Edge 1: 1
3
Edge 2: 1
6
Edge 3: 4
7
Edge 4: 4
2
Edge 5: 7
5
Edge 6: 2
5
Edge 7: 6
3
Edge 8: 5
4
```

```
Adjacency Matrix:
      v0  v1  v2  v3  v4  v5  v6
v0    0   0   0   0   0   0   0
v1    0   0   0   1   0   0   1
v2    0   0   0   0   0   1   0
v3    0   0   0   0   0   0   0
v4    0   0   1   0   0   0   0
v5    0   0   0   0   1   0   0
v6    0   0   0   1   0   0   0

Enter starting vertex for DFS: 1

DFS Traversal starting from vertex 1: 1 3 6
Press any key to continue . . .
```

Date:	
Program 13 (b)	Program to perform Breadth First Traversal (BFS) on a Graph

```
#include <stdio.h>

#define MAX 100


int adj[MAX][MAX];

int visited[MAX];

int vertices;


int queue[MAX];

int front = -1, rear = -1;


void enqueue(int vertex) {

    if (rear == MAX - 1) {

        printf("Queue Overflow\n");

        return;

    }

    if (front == -1)

        front = 0;

    rear++;
```

```
    queue[rear] = vertex;
}

int dequeue() {
    if (front == -1 || front > rear) {
        printf("Queue Underflow\n");
        return -1;
    }
    int vertex = queue[front];
    front++;
    return vertex;
}

int isEmpty() {
    return (front == -1 || front > rear);
}

void BFS(int start) {
    int i;

    for (i = 0; i < vertices; i++) {
        visited[i] = 0;
    }
```

```
    front = rear = -1;

    printf("%d ", start);
    visited[start] = 1;
    enqueue(start);

    while (!isEmpty()) {
        int current = dequeue();

        for (i = 0; i < vertices; i++) {
            if (adj[current][i] == 1 && !visited[i]) {
                printf("%d ", i);
                visited[i] = 1;
                enqueue(i);
            }
        }
    }
}

void displayMatrix() {
    int i, j;
```

```
    printf("Adjacency Matrix:\n");
    printf("     ");
    for (i = 0; i < vertices; i++) {
        printf(" v%d ", i);
    }
    printf("\n");

    for (i = 0; i < vertices; i++) {
        printf("v%d   ", i);
        for (j = 0; j < vertices; j++) {
            printf(" %d  ", adj[i][j]);
        }
        printf("\n");
    }
}

int main() {
    int edges, src, dest, start;
    int i, j;

    printf("Enter the number of vertices (max 100): ");
    scanf("%d", &vertices);
```

```
printf("Enter the number of edges: ");
scanf("%d", &edges);

for (i = 0; i < vertices; i++) {
    for (j = 0; j < vertices; j++) {
        adj[i][j] = 0;
    }
    visited[i] = 0;
}

printf("\nEnter edges (source destination):\n");
for (i = 0; i < edges; i++) {
    printf("Edge %d: ", i + 1);
    scanf("%d %d", &src, &dest);
    adj[src][dest] = 1;
}

displayMatrix();

printf("\nEnter starting vertex for BFS: ");
scanf("%d", &start);

printf("\nBFS Traversal starting from vertex %d: ", start);
```

```
    BFS(start);

    printf("\n");


    return 0;

}
```

```
"D:\DFS Lab Programs\BFS.ex

Enter the number of vertices (max 100): 7
Enter the number of edges: 6

Enter edges (source destination):
Edge 1: 1
3
Edge 2: 4
2
Edge 3: 5
6
Edge 4: 3
4
Edge 5: 5
2
Edge 6: 4
3
```

```
Adjacency Matrix:
      v0  v1  v2  v3  v4  v5  v6
v0    0   0   0   0   0   0   0
v1    0   0   0   1   0   0   0
v2    0   0   0   0   0   0   0
v3    0   0   0   0   1   0   0
v4    0   0   1   1   0   0   0
v5    0   0   1   0   0   0   1
v6    0   0   0   0   0   0   0

Enter starting vertex for BFS: 1

BFS Traversal starting from vertex 1: 1 3 4 2
Press any key to continue . . .
```

Date:	
Program 14	Program to implement B tree and perform operations (Insert,Search,Display)

```
#include <stdio.h>
#include <stdlib.h>

#define T 2

typedef struct BTree
{
   int n;
   int leaf;
   int key[2 * T - 1];
   struct BTree *child[2 * T];
} node;

node *root = NULL;

node* createNode(int leaf)
{
 int i;
   node *newNode = (node*)malloc(sizeof(node));
   newNode->leaf = leaf;
   newNode->n = 0;
   for (i = 0; i < 2 * T; i++)
      newNode->child[i] = NULL;
   return newNode;
}
void traverse(node *ptr)
```

```
{
  if (ptr != NULL)
    {
        int i;
        for (i = 0; i < ptr->n; i++)
        {
            if (!ptr->leaf)
                traverse(ptr->child[i]);
            printf("%d ", ptr->key[i]);
        }
        if (!ptr->leaf)
            traverse(ptr->child[i]);
    }
}
node* search(node *ptr, int k)
{
    int i = 0;
    while (i < ptr->n && k > ptr->key[i])
        i++;
    if (i < ptr->n && k == ptr->key[i])
        return ptr;
    if (ptr->leaf)
        return NULL;
    return search(ptr->child[i], k);
}
void splitChild(node *parent, int i, node *fullChild)
{
 int j;
    node *newChild = createNode(fullChild->leaf);
    newChild->n = T - 1;
    for (j = 0; j < T - 1; j++)
```

```
        newChild->key[j] = fullChild->key[j + T];
    if (!fullChild->leaf)
        {
        for (j = 0; j < T; j++)
            newChild->child[j] = fullChild->child[j + T];
    }
    fullChild->n = T - 1;
    for (j = parent->n; j >= i + 1; j--)
        parent->child[j + 1] = parent->child[j];
    parent->child[i + 1] = newChild;
    for (j = parent->n - 1; j >= i; j--)
        parent->key[j + 1] = parent->key[j];
    parent->key[i] = fullChild->key[T - 1];
    parent->n = parent->n + 1;
}
void insertNonFull(node *ptr, int k)
{
    int i = ptr->n - 1;
    if (ptr->leaf)
    {
        while (i >= 0 && k < ptr->key[i])
        {
            ptr->key[i + 1] = ptr->key[i];
            i--;
        }
        ptr->key[i + 1] = k;
        ptr->n = ptr->n + 1;
    }
    else
    {
        while (i >= 0 && k < ptr->key[i])
```

```
            i--;
        i++;
        if (ptr->child[i]->n == 2 * T - 1)
 {
            splitChild(ptr, i, ptr->child[i]);

            if (k > ptr->key[i])
                i++;
        }
        insertNonFull(ptr->child[i], k);
    }
}
void insert(int k)
{
    if (root == NULL)
    {
        root = createNode(1);
        root->key[0] = k;
        root->n = 1;
        return;
    }
    if (search(root, k) != NULL)
    {
        printf("Duplicate key not allowed.\n");
        return;
    }
    if (root->n == 2 * T - 1)
    {
        node *newRoot = createNode(0);
        newRoot->child[0] = root;
        splitChild(newRoot, 0, root);
```

```
        int i = 0;
        if (k > newRoot->key[0])
            i++;
        insertNonFull(newRoot->child[i], k);
         root = newRoot;
    }
    else
    {
        insertNonFull(root, k);
    }
}
int main()
{
    int ch, val;
    node *found;

    while (1)
    {
        printf("\n--- B-Tree Menu ---\n");
        printf("1. Insert\n");
        printf("2. Search\n");
        printf("3. Traverse\n");
        printf("4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &ch);
        switch (ch)
        {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &val);
                insert(val);
```

```
        break;
    case 2:
        if (root == NULL)
        {
            printf("Tree is empty.\n");
    }
        else
        {
            printf("Enter value to search: ");
            scanf("%d", &val);
            found = search(root, val);

            if (found != NULL)
                printf("%d found in B-Tree.\n", val);
            else
                printf("%d not found in B-Tree.\n", val);
        }
        break;
    case 3:
        if (root == NULL)
            printf("Tree is empty.\n");
        else
        {
            printf("Traversal: ");
            traverse(root);
            printf("\n");
        }
        break;
    case 4:
        printf("Exiting...\n");
        exit(0);
```

```
        default:
            printf("Invalid choice.\n");
    }
  }
  return 0;
}
```

```
"D:\DFS Lab Programs\BTree_

--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 1
Enter value to insert: 64

--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 1
Enter value to insert: 25

--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 1
Enter value to insert: 99
```

```
--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 1
Enter value to insert: 45

--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 1
Enter value to insert: 70

--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 1
Enter value to insert: 52
```

```
--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 3
Traversal: 25 45 52 64 70 99

--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 2
Enter value to search: 52
52 found in B-Tree.
```

```
--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 2
Enter value to search: 100
100 not found in B-Tree.

--- B-Tree Menu ---
1. Insert
2. Search
3. Traverse
4. Exit
Enter choice: 4
Exiting...
Press any key to continue . . .
```

Date:	
Program 15	Program to perform Hashing using Mid-Square Method and Linear Probing (Collision Resolution)

```
#include <stdio.h>

#include <stdlib.h>

#include <math.h>

#define SIZE 10

#define EMPTY -1


int hashTable[SIZE];

int midSquareHash(int key) {

    long long square = (long long)key * key;

    int mid;

    if (square < 10) {

        mid = (int)square;

    } else {

        mid = (int)(square / 10) % 10;

    }

    return mid % SIZE;

}

void insert(int key) {

    int index = midSquareHash(key);
```

```
    int i;
    for (i = 0; i < SIZE; i++) {
        int probeIndex = (index + i) % SIZE;


        if (hashTable[probeIndex] == EMPTY) {
            hashTable[probeIndex] = key;
            printf("Key %d inserted at index %d\n", key, probeIndex);
                return;
        }
    }
    printf("Hash Table is full! Cannot insert %d\n", key);
}
void display() {
    int i;
    printf("\nHash Table Contents:\n");
    printf("Index\tKey\n");
    for (i = 0; i < SIZE; i++) {
        if (hashTable[i] != EMPTY)
            printf("%d\t%d\n", i, hashTable[i]);
        else
            printf("%d\t---\n", i);
    }
}
```

```
int main() {
    int i, choice, key;
    for (i = 0; i < SIZE; i++) {
        hashTable[i] = EMPTY;
    }
    while (1) {
        printf("\n1. Insert Key\n2. Display Table\n3. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter key to insert: ");
                scanf("%d", &key);
                insert(key);
                    break;
            case 2:
                display();
                break;
            case 3:
                exit(0);
```

```
        default:
            printf("Invalid choice!\n");
    }
  }
  return 0;
}
```

```
"D:\DFS Lab Programs\Hashir

1. Insert Key
2. Display Table
3. Exit
Enter choice: 1
Enter key to insert: 15
Key 15 inserted at index 2

1. Insert Key
2. Display Table
3. Exit
Enter choice: 1
Enter key to insert: 65
Key 65 inserted at index 3
```

```
1. Insert Key
2. Display Table
3. Exit
Enter choice: 1
Enter key to insert: 30
Key 30 inserted at index 0

1. Insert Key
2. Display Table
3. Exit
Enter choice: 1
Enter key to insert: 78
Key 78 inserted at index 8
```

```
1. Insert Key
2. Display Table
3. Exit
Enter choice: 2

Hash Table Contents:
Index   Key
0       30
1       ---
2       15
3       65
4       ---
5       ---
6       ---
7       ---
8       78
9       ---

1. Insert Key
2. Display Table
3. Exit
Enter choice: 3
Press any key to continue . . .
```